RooRoom: Ten AI Agents Inside Every UMKC Class

Breaking the barrier between students and professors

Every student can speak up, every professor can listen: AI agents built from LLM loops, over one shared course memory

PRESENTED BY

Bhavana Ediga

[Team member]

Hackaroo '26 · AI Track

1 The problem

Learning happens in one place, but help is scattered across chatbots, email and office hours.

12 am
A general chatbot doesn't know *this* course, and happily does graded homework.

30×
Professors answer the same question one email at a time, days later, in one language.

0
Honest feedback never arrives, because nobody wants to be named.

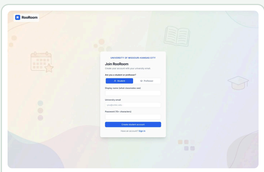
Our idea: break the barrier between students and professors. Put a team of specialised AI agents inside every class. They share one course memory, so **one professor answer reaches every student**, in every language.

2 The product

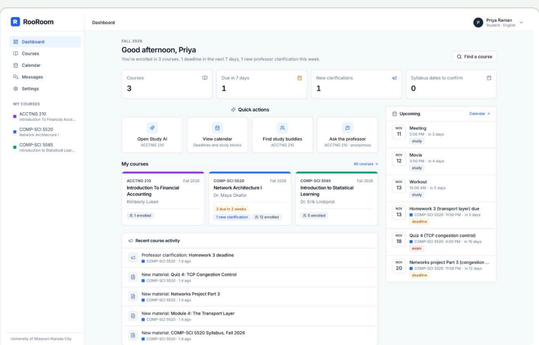
A working web app on real UMKC data: 4,951 courses from the catalog and every Fall 2026 section and professor.



Before sign-in
One scrolling page: every feature and what students can do.



Sign up
Student or professor, @umkc.edu only; professors auto-linked to their courses.



After sign-in: dashboard — courses, deadlines, quick actions
Every number is live: deadlines from the syllabus, new clarifications, and one-click jumps into each agent.

A STUDENT'S JOURNEY

1. **Join** a course: search the UMKC catalog and pick your professor's section.
2. **Study** with the Tutor Agent: answers, summaries and flashcards from course files.
3. **Ask** anonymously; the Filter and Redactor agents clean it before anyone reads it.
4. **Plan** by voice: the Calendar Agent books study time and syllabus deadlines.

WHY NOT JUST A CHATBOT?

What a student needs	General chatbot	RooRoom
Knows this course's files	✗	✓ with source
Hints, not answers, on graded work	✗	✓
Reaches the professor anonymously	✗	✓
One answer reaches the whole class	✗	✓

3 Meet the agents

Each agent is an LLM loop with one job, its own instructions, and only the tools it needs. An orchestrator chains them.

LEARNING

Tutor Agent — RAG · hybrid search
Answers only from this course's files and the professor's answers, and names the file and page. Summaries and interactive flashcards with real formulas.

Integrity Guard — embeddings · guardrail
Spots graded quiz and homework questions by meaning, switches the tutor to hints, and a second check rewrites any worked answer.

ASK THE PROFESSOR

Filter Agent — filtering · NLP
Removes insults and slurs; refuses messages that are only abuse. Labels only, so a message can't talk it into approving itself.

Redactor Agent — NLP
Strips names, IDs and anything that could identify the student.

Digest Agent — clustering
Groups questions by meaning; at 3 students writes the professor a one-line digest.

Summarize Agent — summarization
One click: every question or feedback note becomes "most common themes, with counts" plus next steps.

Answer Drafter — generation
Drafts the reply; the professor edits and publishes it to the whole class.

PLANNING AND EVERYWHERE

Calendar Voice Agent — agent loop · tool calling · voice-to-text
"Block 2–4 on Thursday for Quiz 4." Plans, calls calendar tools, observes, repeats (up to 6 steps); asks before deleting or double-booking.

Syllabus Reader — extraction
Reads a syllabus PDF and sends every deadline to students' calendars.

Translator Agent — NLP
English first; one click translates any answer, keeping formulas intact.

Study buddies are matched from what classmates study with the Tutor and their shared free time.

4 AI inside

AI agents · agent loops · orchestration · tool calling · retrieval (RAG) · embeddings · semantic clustering · content filtering · NLP translation · summarization · voice-to-text · text-to-speech · guardrails · LaTeX math rendering

HOW WE DESIGN OUR AGENTS

One job each
Small agents with narrow instructions are easier to test and harder to misuse.

Least privilege
Only the Calendar Agent has tools, and only on its owner's calendar.

Human in the loop
Professors approve answers; the calendar asks before risky changes.

By the numbers

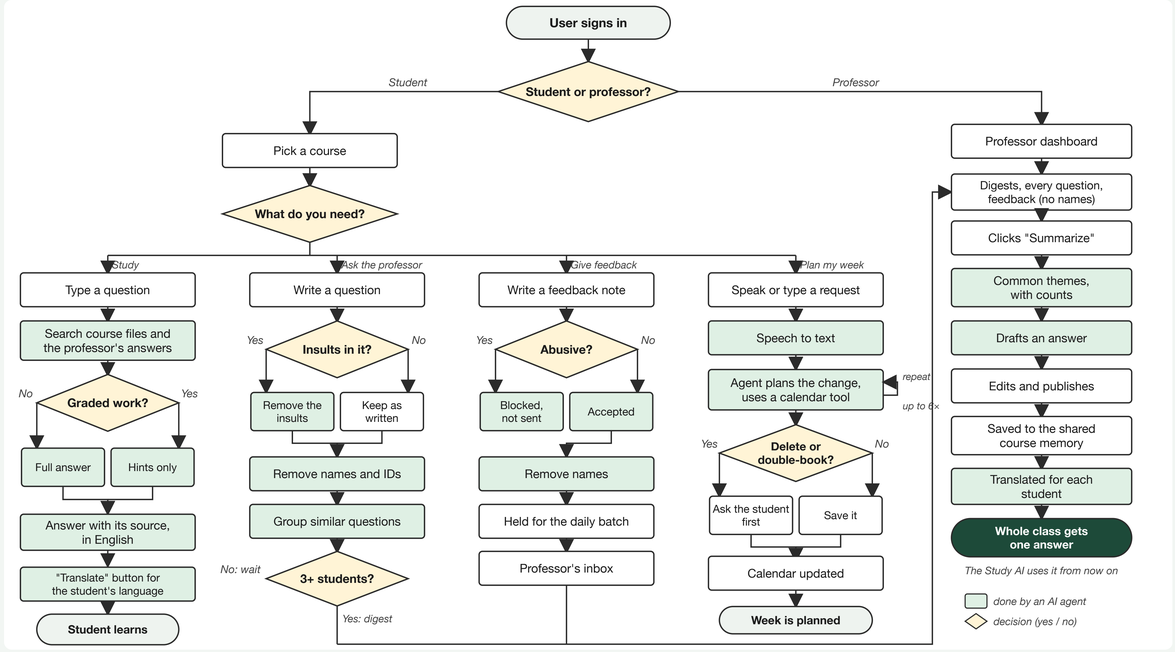
10
AI agents, each with one job

4,951
real UMKC courses, all majors

1,132
Fall 2026 professors to choose from

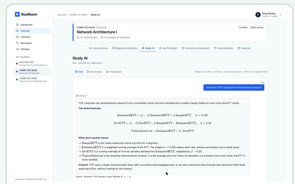
1 click
to summarize every question

5 Workflow

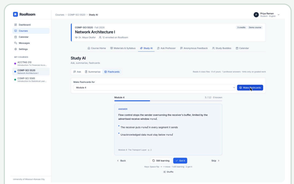


Green steps are done by an AI agent; white steps are the student or professor. The question and feedback paths both end in the professor's view.

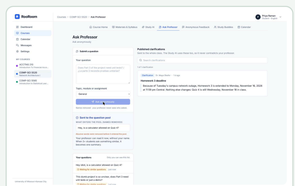
6 The agents at work



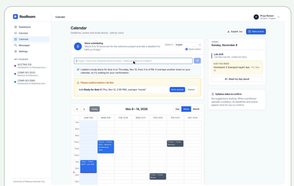
Tutor Agent — formulas typeset
Answers from the right page; one click translates.



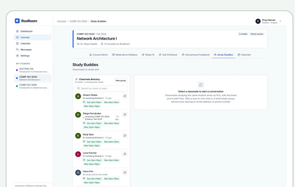
Flashcards — interactive deck
Flip, mark "got it", review what's left.



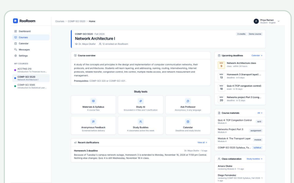
Filter Agent — abuse removed
"Hey idiot, …" reaches the professor as "Hey, …"



Calendar Agent — asks first
Finds the overlap with a movie, asks before booking.

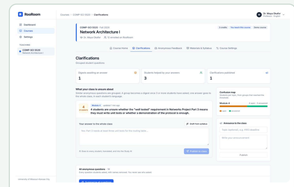


Study Buddies — matched by agents
Same module, times you're both free.

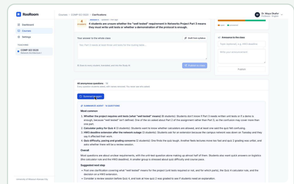


Course page — one hub per class
Deadlines, materials and every agent in one place.

7 The professor's view



Digest Agent — 4 students, one line
Four students asked the same thing in different words; the professor sees one sentence.



Summarize Agent — 14 → themes
"6 students: what does 'well tested' mean?" instead of thirty separate emails.

- ✓ **Read every question**, names removed, never who asked
- ✓ **Digest at 3+ students**, with a drafted answer to edit
- ✓ **Reply to anonymous notes** without ever unmasking
- ✓ **One-click summary** of all questions or all feedback
- ✓ **Publish once**: every student, every language, plus the Tutor
- ✓ **Upload a syllabus**: deadlines reach every calendar

8 Securing the agents

18/18
red-team attacks blocked

0
outside links in AI output

100%
tables under row security

Trust — text is data
Student text and PDFs reach agents as labelled untrusted data, never instructions. Agents that read it have no tools.

Authority — see only what you may
Every agent query runs as the signed-in user under database row rules. No master key.

Egress — nothing leaves
Links and images are stripped from AI output; servers reach only the model endpoint.

Next: one-time email codes (MFA) at sign-in, then a one-course pilot measuring answer quality and professor time saved.